Homework

DM12 - Projection and intersection - 55 min.

In an orthonormal coordinate system of space, consider the point $M(3;0;0)$ and the planes $\mathcal{P} : x + 2y + 2z = 9$ and $\mathcal{Q} : x - y = 0$.

1. Give the coordinates of a nonzero normal vector to $\mathcal{P}$.
2. Determine the coordinates of the orthogonal projection H of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.
3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.
4. The point H lie on this line? Justify.
5. Determine the coordinates of the point K of this line whose z-coordinate is zero. Calculate MK, then compare MK and MH.